

Pairing Proofs and Polynomial Ring Toolkit

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Abstract. In this paper we prove that the One-Time-Pad has perfect security.
[8]

Keywords: Elliptic curve pairings · Zero-knowledge proofs · Polynomial identity testing · Blockchain verification · Recursive proofs.

1 Introduction

Widely used primitives like the AES [?] do not have perfect security, and can be analysed with linear cryptanalysis [?], differential cryptanalysis [?], or differential power analysis [?]. We show that the One-Time-Pad is unconditionally secure in ??.
[9]

2 Main Result